

44 Chapter Review



➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkz9t.

CONCEPT 44.1

Osmoregulation balances the uptake and loss of water and solutes (pp. 978–982)

Animal	Inflow/Outflow	Urine
Freshwater fish. Lives in water less concentrated than body fluids; fish tends to gain water, lose salt	Does not drink water Salt in (active transport by gills) H ₂ O in Salt out H ₂ O out	 ▶ Large volume of urine ▶ Urine is less concentrated than body fluids
Marine bony fish. Lives in water more concentrated than body fluids; fish tends to lose water, gain salt	Drinks water Salt in H ₂ O out Salt out (active transport by gills)	 ▶ Small volume of urine ▶ Urine is slightly less concentrated than body fluids
Terrestrial vertebrate. Terrestrial environment; tends to lose body water to air	Drinks water Salt in (by mouth) H ₂ O and salt out	 ▶ Moderate volume of urine ▶ Urine is more concentrated than body fluids

- Cells balance water gain and loss through **osmoregulation**, a process based on the controlled movement of solutes between internal fluids and the external environment and on the movement of water, which follows by osmosis.
- Osmoconformers** are isoosmotic with their marine environment and do not regulate their **osmolarity**. In contrast, **osmoregulators** control water uptake and loss in a hypoosmotic or hyperosmotic environment, respectively. Water-conserving excretory organs help terrestrial animals avoid desiccation, which can be life-threatening. Animals that live in temporary waters may enter a dormant state called **anhydrobiosis** when their habitats dry up.
- Transport epithelia** contain specialized epithelial cells that control the solute movements required for waste disposal and osmoregulation.

? Under what environmental conditions does water move into a cell by osmosis?

CONCEPT 44.2

An animal's nitrogenous wastes reflect its phylogeny and habitat (pp. 982–983)

- Protein and nucleic acid metabolism generates **ammonia**. Most aquatic animals excrete ammonia. Mammals and most adult amphibians convert ammonia to the less toxic **urea**, which is excreted with a minimal loss of water. Insects and many reptiles, including birds, convert ammonia to **uric acid**, a mostly insoluble waste excreted in a paste-like urine.
- The kind of nitrogenous waste excreted depends on an animal's habitat, whereas the amount excreted is coupled to the animal's energy budget and dietary protein intake.

MAKE CONNECTIONS Metabolism of carbohydrates and fats requires several nitrogen-containing molecules, such as NAD⁺/NADH (see Figure 9.12), but is not a significant source of nitrogen-containing waste. Why?

CONCEPT 44.3

Diverse excretory systems are variations on a tubular theme (pp. 983–987)

- Most excretory systems carry out **filtration**, **reabsorption**, **secretion**, and **excretion**. Invertebrate excretory systems include the **protonephridia** of flatworms, the **metanephridia** of earthworms, and the **Malpighian tubules** of insects. **Kidneys** function in both excretion and osmoregulation in vertebrates.
- Excretory tubules (consisting of **nephrons** and **collecting ducts**) and blood vessels pack the mammalian kidney. Blood pressure forces fluid from blood in the **glomerulus** into the lumen of **Bowman's capsule**. Following reabsorption and secretion, **filtrate** flows into a collecting duct. The **ureter** conveys urine from the **renal pelvis** to the **urinary bladder**.

? What is the function of the filtration step in excretory systems?

CONCEPT 44.4

The nephron is organized for stepwise processing of blood filtrate (pp. 987–993)

- Within the nephron, selective secretion and reabsorption in the **proximal tubule** alter filtrate volume and composition. The descending limb of the **loop of Henle** is permeable to water but not salt; water moves by osmosis into the interstitial fluid. The ascending limb is permeable to salt but not water; salt leaves by diffusion and by active transport. The **distal tubule** and collecting duct regulate K⁺ and NaCl levels in body fluids.
- In mammals, a **countercurrent multiplier system** involving the loop of Henle maintains the gradient of salt concentration in the kidney interior. Urea exiting the collecting duct contributes to the osmotic gradient of the kidney.
- Natural selection has shaped the form and function of nephrons in various vertebrates to the osmoregulatory challenges of the animals' habitats. For example, desert mammals, which excrete the most hyperosmotic urine, have loops of Henle that extend deep into the **renal medulla**, whereas mammals in moist habitats have shorter loops and excrete more dilute urine.

? How do cortical and juxtamedullary nephrons differ with respect to reabsorbing nutrients and concentrating urine?